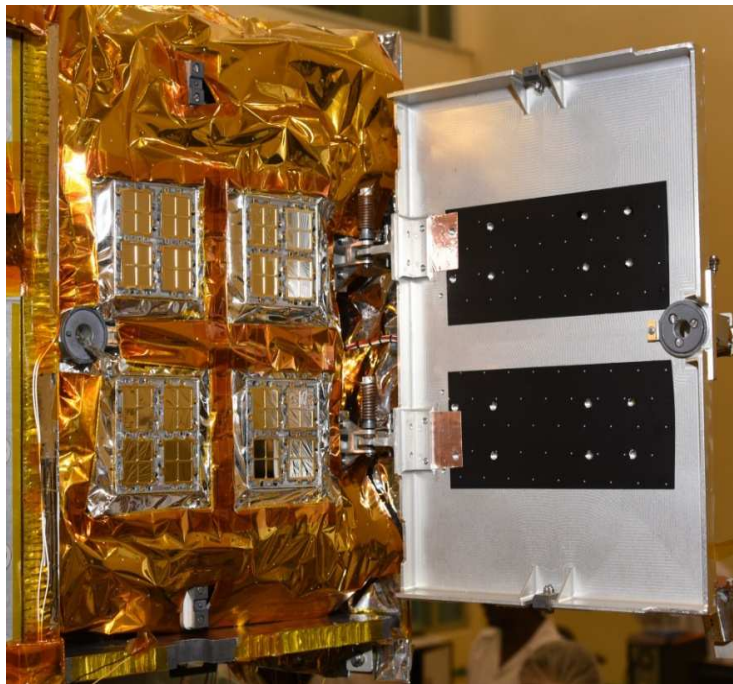


CLASS USER MANUAL

X-ray Fluorescence Spectrometer on Chandrayaan-2 orbiter



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VERSION: 1.1

CLASS is an X-ray fluorescence spectrometer onboard the Chandrayaan-2 Orbiter. This document describes the analysis and software tools with the uncertainties involved. This is intended to serve as a manual for users of CLASS data publicly available through PRADAN web site hosted at ISSDC

Version change history

Version number	Date	Revision description
1.0	28.12.2020	First version in the public archive
1.1	30.12.2021	Updates on number of detectors used and corresponding Ancillary Response File (ARF)

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Table 1 Abbreviations-

CLASS	Chandrayaan-2 Large Area Soft x-ray Spectrometer
DD	Dawn-Dusk
IDL	Interactive Data Language
ISSDC	Indian Space Science Data Centre
NM	Noon-Midnight
PDS4	Planetary Data System (Archival standard)
NM	Noon-Midnight
SCD	Swept Charge Device
UTC	Universal Time
XRF	X-ray Fluorescence
XSPEC	X ray Spectral Package

Introduction

Chandrayaan-2 Large Area Soft x-ray Spectrometer (CLASS) is an X ray fluorescence (XRF) experiment on the Chandrayaan-2 Orbiter to map the surface elemental abundances. Atoms on the lunar surface emit XRF photons when solar X-rays interact with them. CLASS measures the energy and number of these photons from which the elemental abundances corresponding to the element can be determined. This document describes the steps involved in analysing the L1 data product that would be available from the PRADAN interface at ISSDC, ISRO.

Before going any further, please note the following

This is low level (Level 1) data and is made available in the public domain on the best effort basis. It is meant only for those advanced users who can model the lunar X-ray fluorescence spectra (which include calculation of the fluxes of the expected characteristic lines of various constituent elements as well as the scattered solar X-ray spectrum) from the lunar surface for given solar X-ray spectra and observation geometry. The necessary details of the observation geometry for each file are available in the header of the Level 1 data file. Users need to fit the observed spectra with the calculated spectra using suitable spectral fitting software (suggested software is XSPEC, which is available as a part of HEASoft). Other files required for analysis (.arf, .rmf and a generic instrument background) are also made available. In case of low solar X-ray activity or smaller flares, it is necessary add up multiple fits files.

For the users interested only in the elemental abundance values, it is recommended to use higher level data (Level 2), that will be released subsequent to this release.

Instrument

CLASS consists of sixteen swept charge devices (SCDs) numbered 0 to 15. Figure 1 shows the placement of the SCDs in the orientation its facing the lunar surface. There is a Cu collimator coated with gold above each SCD that defines the field of view which translates to an instantaneous ground foot print of 12.5 km x 12.5 km. For an interval of 8s, the entire IFOV shifts by 12.5 km along the track. Even though the overall size of the footprint for 8s duration is 25 km x 12.5 km, 75 % of the counts arise from the central 12.5 km x 12.5 km which is taken as the FWHM spatial resolution.

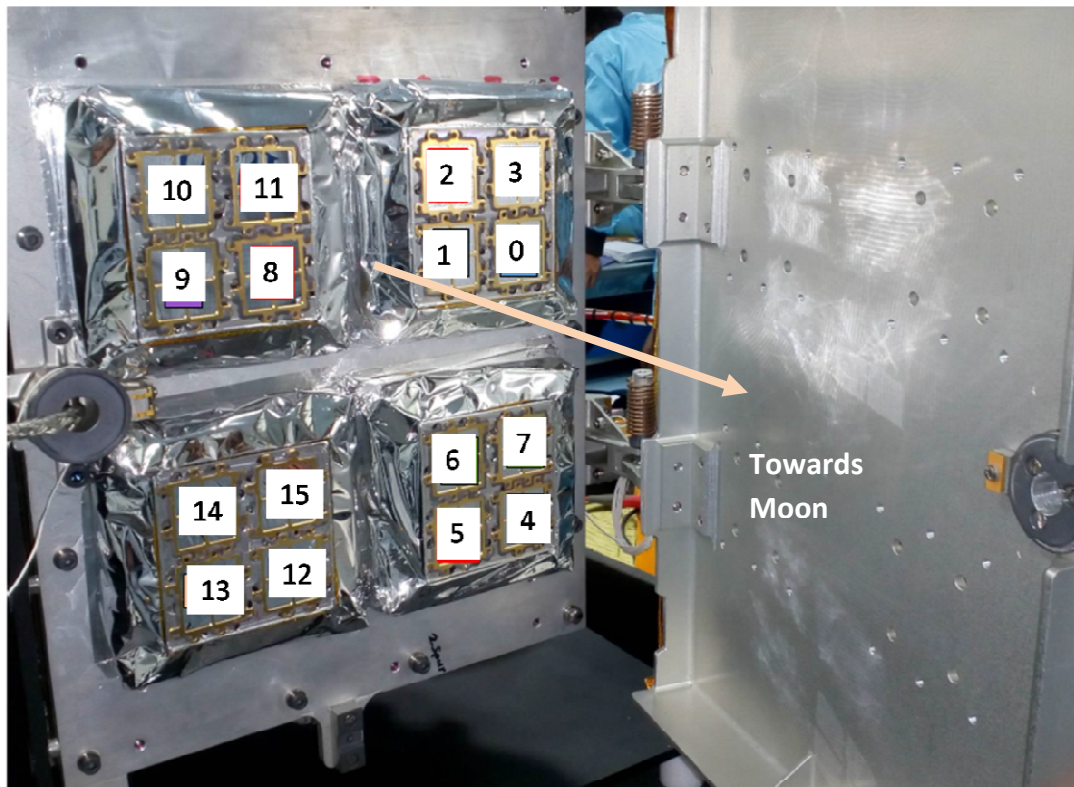


Figure 1: CLASS spectrometer with 16 detectors numbered

Al filters are used to block visible light. Figure 2 shows a cross sectional view of one of the SCDs with collimator and Al filters. Each quad consisting of 4 SCDs have a thermistor that records the operating temperature every 64ms and is sent as part of telemetry.

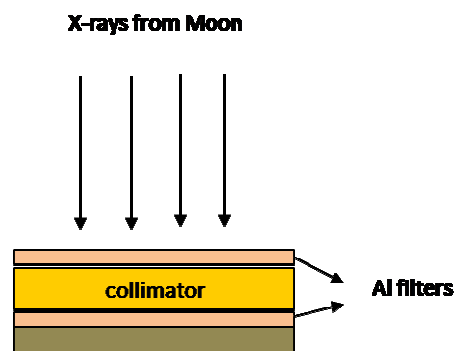


Figure 2: Schematic cross sectional view of the path of X ray photons reaching the SCD

There is a light leakage into four of the SCDs (12, 13, 14, 15) when the Sun is at grazing angles with respect to the plane of SCD array. This happens at high latitudes at both poles. In the first version of the release we used 12 of the SCDs (0-11) to generate the L1 product. In the second version of the data and all subsequent releases we have ensured maximal utilisation of the data. This is done by selectively adding SCDs without contamination from visible light for all latitudes to generate the L1 product.

CLASS is operating in the sunlit side of the Chandrayaan-2 polar circular orbit as shown in Figure 3. During Noon-Midnight season (when the Sun vector is parallel to the plane of the orbit), CLASS operates 59 minutes every orbit while in the Dawn-Dusk season (when the Sun vector is perpendicular to the plane of the orbit), it is for the whole orbit (118 minutes). However lunar XRF is detected only when there is a solar flare.

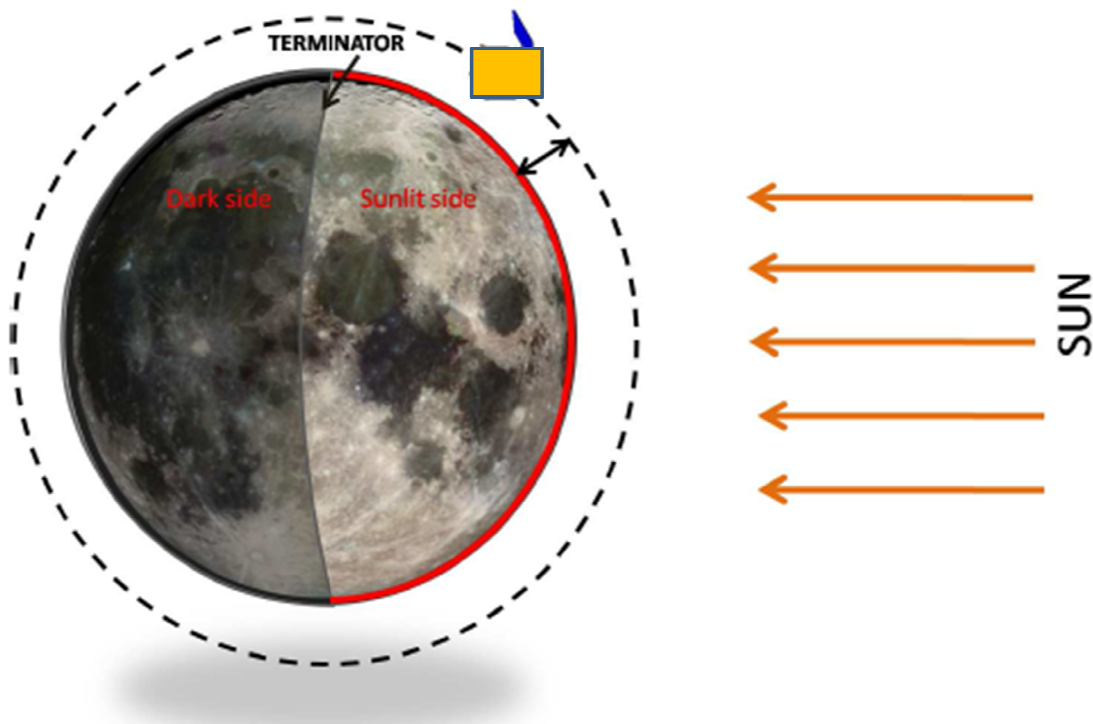


Figure 3: CLASS operational scenario in the polar circular orbit of Chandrayaan-2 Spacecraft during N-M geometry

Further reading:

1. Chandrayaan-2 Large Area Soft x-ray Spectrometer, Radhakrishna V, Anurag Tyagi, S. Narendranath, Koushal Vadodariya, Reena Yadav, Brajpal Singh, G Balaji, Neeraj Satya, Akash Shetty, H.N. Suresha Kumar, Kumar, Vaishali S, Netra S Pillai, S. Tadeipalli, Venkata Raghavendra, P. Sreekumar, Anil Agarwal, Valarmathi N, Current Science, Jan, 2020.
2. Chandrayaan-2 Large Area Soft x-ray Spectrometer: Calibration, Inflight performance and first results, Netra S Pillai, S. Narendranath, K. Vadodariya, S. Tadeipalli, Radhakrishna V, Anurag Tyagi, Reena Yadav, Brajpal Singh, Vaishali S, P. S. Athiray, P. Sreekumar, K. Sankarasubramanian, Amit Basu Sarbhadhikari, Megha Bhatt, Mithun N P S, Santosh Vadawale, *Icarus*, 2021.

CLASS Data

CLASS raw data is processed at the Payload Operations Centre (POC) into L1 and L2 products for archival at ISSDC. The data products, formats and file naming conventions are described in detail in the “SIS document”. In brief,

Data product	Description	Suggested quick look tool
L1	Combined spectrum for 8 s	FITS VIEWER (fv) or ds9
L2	Elemental map	QGIS (or ArcGIS)

The meta data along with the data files contain all necessary information such as UTC, lunar co-ordinates, geometry of observation and file history that is necessary for understanding the data set. L1 data is in FITS format that can be read in Matlab, IDL, Python or any such programming language. We use FITSVIEWER that is available as part of FTOOLS (or standalone) from https://fits.gsfc.nasa.gov/fits_viewer.html. This is further described in the section ‘Analysis’.

For L1 data sets, we suggest that the user selects data during a solar flare by identifying flare times from solar observatories such as GOES. For a solar flare that spans the duration, for example 10.00 UT to 10.10 UT, download data with an additional duration of 10 minutes on either side for estimating background.

How to download data

L1 data during operations (irrespective of XRF signal) is available for download from PRADAN (<https://pradan.issdc.gov.in/pradan/>), the data dissemination website for Chandrayaan-2 data. All the available calibrated data - combined 8s FITS files - are listed in PRADAN and user can select the required files and download as a zip file. The downloaded zip file will have the naming convention for eg. cla_2020Dec10T051032456.zip. The unzipped data dir will be as follows

```
cla/data/calibrated/yyyy/mm/dd/  
eg. cla/data/calibrated/2019/10/29/
```

Any further data downloads will unzip into cla/data/calibrated/yyyy/mm/dd.

After downloading data from PRADAN, a zip file containing the calibration files, documents and analysis software should be downloaded from the ‘Other Downloads’ page under ‘CLASS’ section in PRADAN website.

Once unzipped, the main directory 'cla' contains the following directories

- calibration
- document
- miscellaneous

'calibration' directory contains the files required to carry out analysis.

1. class_rmf_v1.rmf
2. class_arf_v1.arf
3. background_all_events.fits

'document' directory contains this user manual and SIS document which provides details of the file naming conventions and formats

'miscellaneous' directory contains a zip file which provides a software program (CLASS_add_L1_files_time.pro) written in IDL, to add the 8s files.

Once the above mentioned files are downloaded, users can carry out analysis of CLASS calibrated data as explained in the next sections.

Analysis

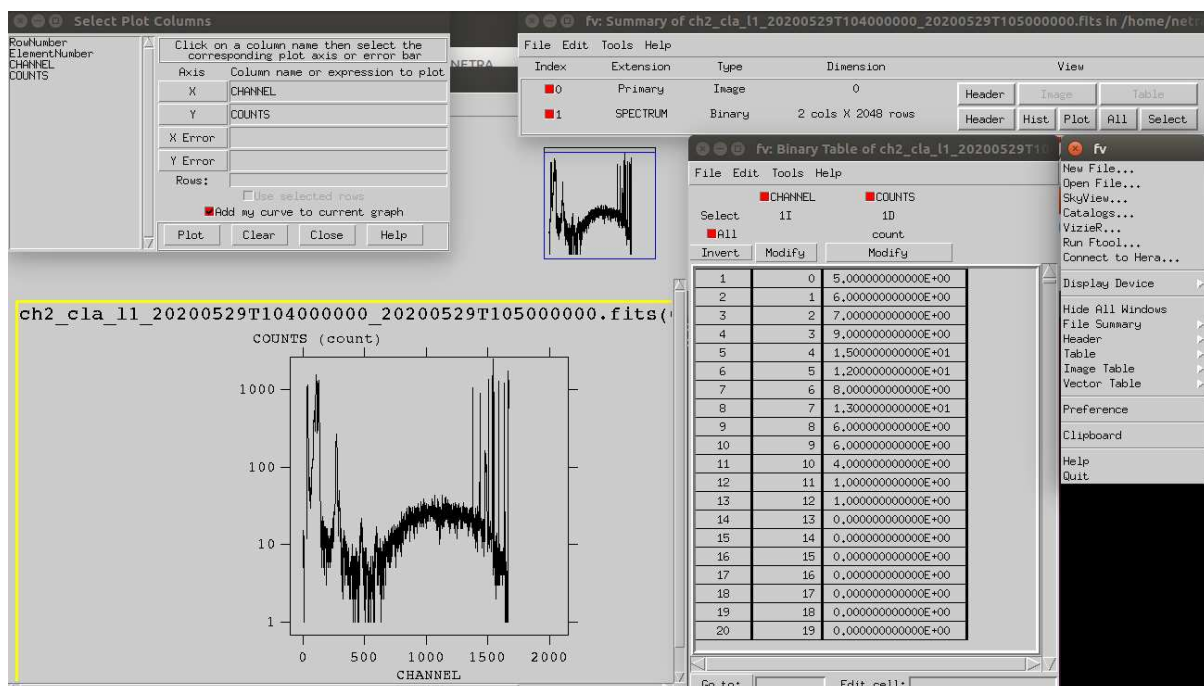
Quick look

The 8 s files are named as

ch2_cla_l1_yyyymmddTHHMMSS_yyyymmddTHHMMSS.fits with the start and end times of the 8 s mentioned in the file name.

```
> fv ch2_cla_l1_20200529T104532257_20200529T104540257.fits
```

The 2 x 2048 data has channel number in the first column and counts in the second column. The tool can be used to plot the spectrum as shown below.



The header of the 8s fits files contains various parameters which are required for higher level analysis. These are described in the SIS document.

Adding Spectra in time

In a spectrum of 8s duration, XRF lines may or may not be detected depending on the strength of the solar flare, season (NM/DD), geometry and composition. The spectra may be added to improve statistics. This is a simple addition in time and users can write their own code or also used 'addpha'

tool in FTOOLS. However, we have provided a tool in IDL, *CLASS_add_L1_files_time.pro* for the users on a best effort basis. It takes start UTC of time to be added and end UTC of the time to be added as inputs and generates the added spectrum. It also generates the requisite header parameters by appropriate averaging. Sample usage of the same is given below.

IDL> CLASS_add_L1_files_time, ip_directory, start.UTC, end.UTC, where

ip_directory = Unzipped L1 directory where 8s fits files are stored

start.UTC = Start time of spectrum to be added

end.UTC = End time of spectrum to be added

If the user wish to add the spectra based on the latitude, the lunar latitude and longitude are provided in the header of every L1 data file. The corresponding time of the observation should be noted and the above steps done to add spectra.

Background

The average background in CLASS arising from ambient particles is given as a background file for analysis. There are also two lines at Al-K α and Cu-K α arising from the instrument that are always present. During the passage through geotail, the particle population is highly dynamic and specific intervals have to be chosen by the user. The counts in the 7-16 keV band may be used as an indicator of changes in particle flux during times of low solar activity. The user must look through the non-flare spectra and generate a background file. A nominal background spectrum from night side observations is provided. However we recommend that the user create specific background ideal for the time of interest. This is especially important for data during geotail passes (+/- 3 days of the full Moon) when the particle background is dynamic.

Response matrices

To analyse the X ray spectrum, a re-distribution matrix file (RMF) and the ancillary response file (ARF) are provided in the CALDB directory. The RMF is a 2048 x 2048 matrix with a Non-Gaussian spectral re-distribution function (SRF). The ARF is a 2 x 2048 matrix that gives the effective area as a function of energy. The data with XRF signals (which are of primary interest to the user) are L1 products with all 16 SCDs in the ~ -88 to +88 range of latitudes. Hence we provide an ARF with 16 SCDs to the user.

CLASS is operating most of the time well below -30 deg C in orbit and hence a single RMF file caters to the entire data set. For specific periods where temperature is higher we will provide another version of RMF in future releases.

The response matrices *class_rmf_v1.rmf* and *class_arf_v1.arf* have to be separately downloaded from the 'other downloads' section of PRADAN web page and placed in the working directory. The response matrix files are in the OGIP standard format readable with a FITS Viewer (https://fits.gsfc.nasa.gov/fits_viewer.html) as well as in the X ray spectral analysis software XSPEC. Detailed descriptions of the format are available at the link below

https://heasarc.gsfc.nasa.gov/docs/heasarc/ofwg/docs/spectra/ogip_92_007/node5.html

Spectral analysis

We recommend using XSPEC (<https://heasarc.gsfc.nasa.gov/xanadu/xspec/>) which is an open source tool for X ray spectral analysis. However, the files are in standard FITS formats and can be read by any program written by the user. The instructions to download and install XSPEC are given in the above link. Further spectral analysis steps are given below

```
XSPEC version: 12.10.1f
Build Date/Time: Mon Oct 7 21:57:43 2019

XSPEC12>data ch2_cla_l1_single_20200529T104000000_20200529T105000000.fits
***Warning: Detected response matrix energy bin value = 0 (or neg).
      XSPEC will instead use small finite value (response file will not be altered).
***Warning: No TLMIN keyword value for response matrix FCHAN column.
      Will assume TLMIN = 1.
Warning: RMF TELESCOPE keyword is not consistent with spectrum

1 spectrum in use

Spectral Data File: ch2_cla_l1_single_20200529T104000000_20200529T105000000.fits Spectrum 1
Net count rate (cts/s) for Spectrum:1 1.617e+02 +/- 5.191e-01
Assigned to Data Group 1 and Plot Group 1
Noticed Channels: 1-2048
Telescope: CHANDRAYAAN-2 Instrument: CLASS Channel Type: PHA
Exposure Time: 600 sec
Using fit statistic: chi
Using test statistic: chi
Using Response (RMF) File class_v1.rmf for Source 1
Using Auxiliary Response (ARF) File class_v1.arf

XSPEC12>cpd /xw
XSPEC12>setplot energy
XSPEC12>ignore 0.0-0.4 10.0-**
      30 channels (1-30) ignored in spectrum # 1
      1308 channels (741-2048) ignored in spectrum # 1

XSPEC12>plot ldata
XSPEC12>
```

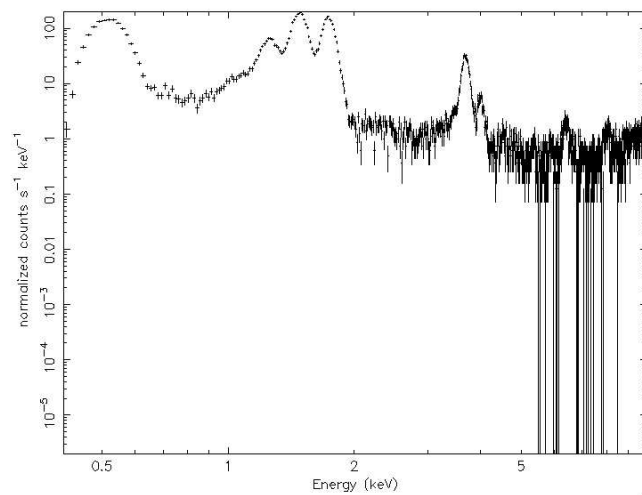


Figure 4: CLASS lunar XRF spectrum plotted in XSPEC

Steps:

- Use a background file : pre/post flare spectrum
- Models: (a) x2abund as a local model (b) scattered component as a table model
- XSPEC fitting routines are used and the best fit gives the elemental wt %.

Calculating the scattered solar spectrum

The solar spectrum for the time interval of interest can be downloaded from the same website from the Solar X-ray Monitor (XSM) on Chandrayaan-2. For times when simultaneous XSM observations are not available, solar spectrum from other solar observatories are sought. We currently use GOES broad band flux to estimate the solar spectrum when XSM is not available. We compute the scattered fraction (elastic scattering) using the solar spectrum and model the scattered fraction observed from the Moon. The scattered component is modelled as a table model in XSPEC. The solar spectrum from XSM is modelled to calculate the scattered component and is then incorporated as a user model (table model) in XSPEC. The scattered model has to be computed from the solar spectrum for the same time interval that is being analysed. The users should either calculate this or fit the component with a continuum + solar lines with knowledge of the lines in the solar X-ray spectrum.

L2- Elemental abundances

The L2 products that would be released are elemental maps for the flares observed during the mentioned period. The elemental abundances are determined by fitting the observed lunar X ray spectrum with a XRF model calculated using a code x2abundance (Athiray et al, 2013a,2013b) that follows the XRF calculation in Shirawai and Fujino (1966). This is implemented as a local model that can be called through pyxspect (Python interface to XSPEC).

Ref:

Shirawai and Fujino, Theoretical calculation of X-ray fluorescence intensities in fluorescent spectrochemical analysis, Japanese journal of applied physics, 1966, 5, 886.

Athiray P. S. et al, Validation methodology to derive elemental abundances from X ray observations on Chandrayaan-1, 2013a, PSS, 75,188-194.

Athiray P. S. et al, Experimental validation of inversion code for Chandrayaan-1, 2-13b, PSS, 89, 183-187.

Points to be noted

- The number of channels is 2048 with a gain of 13.5 eV/channel. The raw spectra are shifted to the 0th channel, hence there is no offset.

$$\text{Energy} = \text{channel number} \times 13.5 \text{ eV}$$

- For lunar spectral analysis use channels in the range 37-800 (0.5-10 keV)
- For any all science analysis ignore channels above 1112 (~ 15 keV) and below 37 (0.5 keV)
- Lone data points (or spikes) in any channel should not be considered for analysis
- Users may rebin the spectra to improve statistics from 2048 channels to 1024. A re-binned rmf will be provided in future releases.

Contacts

Data analysis related queries: kcshyama@ursc.gov.in, netra@ursc.gov.in

Data download, software related queries: vaishali@ursc.gov.in

Appendix 1: L0 to L1

The SCDs in CLASS record each X ray event with a time tag. The raw data received is thus a sequence of time tagged events. Given the 12.5 km ground pixel (FWHM) defined by the mechanical collimator on CLASS, we generate 8 s spectra (it takes 8 seconds to cover 12.5 km) from the time tagged events.

To make a combined spectrum, the counts from all good detectors have to be added. Since the gains are slightly different amongst the devices, the spectra are added in energy space. Each of the SCD is calibrated on ground and the gain at temperatures in the range -55°C to 5°C is measured. A look up table is generated using this. For every 8s spectrum from a single SCD in channel space, the gain corresponding to the operating temperature is read from the look up table. The spectrum from each SCD is converted to energy space. We then rebin the spectra into 2048 channel with 13.5 eV/bin width. The added spectrum is in channel space with a gain of 13.5 eV/channel. There is no offset as all spectra are shifted to begin at channel 0 before this. The combined 8 s spectrum from CLASS is the data set for the user.

From -55°C to $\sim -30^{\circ}\text{C}$, a single RMF can be used. The ARF file is generated for 16 SCDs and remains the same for all data sets. The calibration of CLASS is described in the paper Netra et al, listed in the section 'Further Reading'.

Figure 5: Onboard source spectrum from the combined X ray detectors on CLASS at -53°C

